# Est. Sept 23

miércoles, septiembre 23, 2026 2:12 PM

No está definido.

Pares ordenados: $A \times B = \{(a,b) : a \in A, b \in B\} \to$ producto.

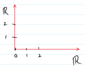
$A = \{1,2,3\}$ $B = \{3,4\}$

$A \times B = \{(1,3), (1,4), (2,3), (2,4), (3,3), (3,4)\}$

$|A \times B| = 6 = |A| \cdot |B| = 3 \cdot 2$

$(a,b) \neq (b,a)$ pares ordenados.

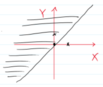
$\mathbb{R}^2 = \{(a,b) : a, b \in \mathbb{R}\}$



$A \subseteq \mathbb{R}^2$

$\hookrightarrow A = \{(x,y) : x \leq y\}$ pinto.

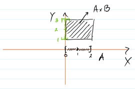
$y = x$



$A = [0,2]$ $B = [1,3]$

$A, B \subseteq \mathbb{R}$

$A \times B \subseteq \mathbb{R} \times \mathbb{R}$



Sea $(a,b) \in A \times B \Rightarrow a \in A, b \in B \Rightarrow a \in \mathbb{R}, b \in \mathbb{R} \Rightarrow (a,b) \in \mathbb{R} \times \mathbb{R}$.

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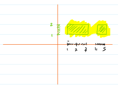
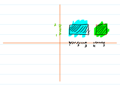
$A = [1,3] \cup [4,5]$ $B = [1,2]$.

$A \times B = \{[1,3] \cup [4,5]\} \times [1,2] \overset{?}{=} [1,3] \times [1,2] \cup [4,5] \times [1,2]$





Teorema: $A, B, C$ conjuntos:

$$(A \cup B) \times C = (A \times C) \cup (B \times C)$$

prueba:

$\supseteq, \subseteq$ Sea $(x,y) \in (A \cup B) \times C \Leftrightarrow x \in A \cup B \wedge y \in C$

$\Leftrightarrow (x \in A \vee x \in B) \wedge (y \in C)$

$\Leftrightarrow (x \in A \wedge y \in C) \vee (x \in B \wedge y \in C)$

$\Leftrightarrow (x,y) \in A \times C \vee (x,y) \in B \times C$

$\Leftrightarrow (x,y) \in (A \times C) \cup (B \times C$. Q.E.D.

Teo. $A \times \emptyset$. $\to \emptyset$

$\to$ No es $\emptyset$.

Supongamos que $A \times \emptyset$ No es $\emptyset$

$\Rightarrow$ Sea $(x,y) \in A \times \emptyset \Rightarrow y \in \emptyset$ ↯

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$A_1 \times A_2 \to A_1 \times A_2 \times A_3 := (A_1 \times A_2) \times A_3$

Tiene sentido $A_1 \times A_2 \times \dots \times A_n$

Df: $(a,b) = ?$ ← ¿cómo lo defino?

$(a,b)$ ~~$= \{a,b\}$~~ $\to$ tengo problemas $\{a,b\} = \{b,a\}$

$(a,b) := \{\{a\}, \{a,b\}\}$

Ej: $(4,6) = \{\{4\}, \{4,6\}\}$

$(6,4) = \{\{6\}, \{6,4\}\}$

¿Son iguales? No: $\{4\} \in$

$\{4\} \notin$

$(4,4) = \{\{4\}, \{4,4\}\} = \{\{4\}, \{4\}\} = \{\{4\}\}$.

$(4,6) \cup (6,4) = \{$~~$\{4\}$~~$,$ ~~$\{4,6\}$~~$\} \cup \{$~~$\{6\}$~~$, \{4,6\}\}$

$= \{\{4\}, \{6\}, \{4,6\}\} = ? \neq (?,?)$

Teorema: $(a,b) = (a',b') \Leftrightarrow a = a' \wedge b = b'$

prueba: Caso 1: $a = b \Rightarrow (a,b) = \{\{a\}\} = \{\{a'\}, \{a',b'\}\}$

$\Rightarrow \{a'\} = \{a', b'\} \Rightarrow a' = b'$

$\{\{a\}\} = \{\{a'\}\} \Rightarrow \{a\} = \{a'\}$

$\Rightarrow a = a' = b = b'$

Caso 2: $a \neq b \Rightarrow (a,b) = \{\{a\}, \{a,b\}\}$ tiene dos elementos.

$\{\{a\}, \{a,b\}\} = \{\{a'\}, \{a',b'\}\}$

Dos casos: $\{a\} = \{a',b'\} \Rightarrow a' = b'$

Una contradiccion porque tiene dos elementos

$\{a\} = \{a'\} \Rightarrow \{a,b\} = \{a',b'\}$.

$b \in \{a,b\} \Rightarrow b \in \{a',b'\}$, $b \neq a'$

porque $a' = a \Rightarrow b = b'$.

$\Leftarrow$ Si $a = a' \wedge b = b' \Rightarrow (a,b) = (a',b')$ Q.E.D.

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3- tupla: $A \times B \times C = (A \times B) \times C$

$(a,b,c) = ((a,b), c) = \{\{(a,b)\}, \{(a,b), c\}\}$

~~$= \{\{\{\{a\}, \{a,b\}\}\}, \{\{a\}, \{a,b\}\}, \{\{\{a\}, \{a,b\}\}, c\}\}$~~

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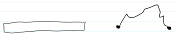
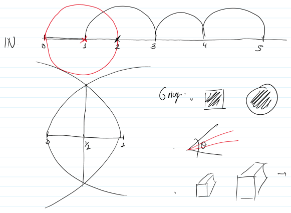
Diosa 1 Diosa 2.

$\to$ volumen doble.

Regla y Compas:

Sin marcas Sin memoria





Griego:

$\to \sqrt[3]{2}$

$A_1 \times A_2 \times A_3 \times A_4 \to [(A_1 \times A_2) \times A_3] \times A_4$

$([(a,b), c], d)$

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$H_a = \{(x,y) : y = a\}$ $H_3$

$V_b = \{(x,y) : x = b\}$ $V_5$